

Amy Hu

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EDUCATION

Bachelor of Science in Mathematical Sciences - Discrete Mathematics & Logic; Additional Major in Computer Science

Aug 2025 - May 2029

Carnegie Mellon University

- Selected coursework: Great Ideas in Theoretical Computer Science, Functional Programming, Probability and Computing, Principles of Imperative Computation, Fundamentals of Programming & Computer Science, Concepts of Mathematics, Discrete Mathematics, Calculus in Three Dimensions, Matrices & Linear Transformations.
- Activities: ScottyLabs Development Team; Badminton Club.

EXPERIENCE

Semisight

May 2026 - Aug 2026

Software Engineering Intern

- Built C++ tools to read and analyze large circuit graphs stored in CSR-style sparse vector formats.
- Implemented fanin/fanout graph traversal to identify fanin cones, loop structures, and re-convergence points.
- Generated structured output files for graph partitions, loop detection, re-convergence, and suppression results.

Carnegie Mellon University

Jun 2026 - Aug 2026

Teaching Assistant - 15-110 Principles of Computing

- Supported introductory computing students through office hours, debugging help, and clear explanations of programming concepts.
- Assisted with grading, student questions, and course logistics for a beginner-focused computer science course.

Mathnasium

Jan 2025 - Jun 2025

Instructor

- Tutored students in mathematics, supported homework and test preparation, and explained new material across skill levels.
- Completed 150+ hours of paid instruction.

Private Tutor

Jun 2023 - Jun 2024

- Provided one-on-one mathematics tutoring for a high school student, supporting assignments and exam preparation.

TECHNICAL SKILLS

Programming: C++, C, Java, Python, JavaScript

Technologies: React, Git/GitHub, Linux, LaTeX, Fusion 360, OpenAI Codex, ChatGPT, LLM-assisted development

Languages: English, Chinese (Simplified/Traditional)

PROJECTS & RESEARCH

UCSD Research - Applied Mathematics Research Assistant

Jun 2024 - Dec 2025

- Conducted applied mathematics research on reconstructing Minkowski geometry from causal separations.
- Contributed to work published in the Journal of Mathematical Physics and participated in weekly research meetings.

USC Biomechanics Project

Jun 2023 - Dec 2024

- Collaborated with the USC Biomechanics Research Lab to develop Python-based vector overlays for visualizing athlete movement.
- Created technical showcase materials, including a body of movement video; completed CITI research certification.

LEADERSHIP & ACTIVITIES

CMU Pre-College

Jun 2026 - Aug 2026

Resident Advisor

- Supported residential community building, event programming, and day-to-day guidance for pre-college students.
- Coordinated with residential staff to maintain a safe, organized, and inclusive student environment.

FIRST Robotics

2022 - 2025

Software, Mentoring & Outreach

- Worked on mechanism-control software and taught Intro to Robot Control during pre-season training.
- Led Arduino/STEM outreach workshops and contributed 100+ volunteer hours supporting younger teams.

Varsity Badminton / Badminton Club

2022 - 2025

Captain; Co-Founder & President

- Led team practices and club activities while expanding off-season training opportunities for teammates.

AWARDS

AP Scholar with Distinction | Presidential Volunteer Service Award (Gold) | FIRST Engineering Inspiration Award | FIRST Impact Award